

UNIVERSITY OF ASIA PACIFIC
DEPARTMENT OF CSE

Lab Report - 8

COURSE CODE: CSE 402.

COURSE TITLE: OPERATING SYSTEM LAB.

SUBMITTED BY:

NAME : SYED MAHATASIN HUSSAIN.

STUDENT ID : 23101140

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SUBMITTED TO

ATIA RAHMAN ORTHI

LECTURER AT UAP

1. Problem:

Given the following reference string and frame size, implement the FIFO (First In First Out) Page Replacement Algorithm using Python.

- **Reference String**: 1, 2, 3, 4, 2, 3, 1
- **Frame Size: 3**

The program should check whether each page is already present in the available frames. If the page is present, count it as a page hit. Otherwise, count it as a page fault and load the page into the frame. When all frames are full, replace the page that entered the frames first.

Finally, calculate the total number of page hits, page faults, hit ratio, and fault ratio.

2. Source Code:

```
# Page Replacement Algorithm (FIFO)

reference = [1, 2, 3, 4, 2, 3, 1]
frame_size = 3

frame = []
hit = 0
fault = 0

for page in reference:

    if page in frame:
        hit += 1
        print(page, "->", frame, "Hit")

    else:
        fault += 1
```

```
        if len(frame) < frame_size:
            frame.append(page)
        else:
            frame.pop(0)
            frame.append(page)

        print(page, "->", frame, "Fault")


hit_ratio = (hit / len(reference)) * 100
fault_ratio = (fault / len(reference)) * 100

print("Total Hit =", hit)
print("Total Fault =", fault)
print("Hit Ratio =", round(hit_ratio,2),"%")
print("Fault Ratio =", round(fault_ratio, 2),"%")
```

3. Output:

```
Problems 2  Output  Debug Console  Terminal  Ports  Spell Checker 2

PS C:\Users\hp\Desktop\CSE 402 Lab> & "C:\Program Files\Python313\python.
1 -> [1] Fault
2 -> [1, 2] Fault
3 -> [1, 2, 3] Fault
4 -> [2, 3, 4] Fault
2 -> [2, 3, 4] Hit
3 -> [2, 3, 4] Hit
1 -> [3, 4, 1] Fault
Total Hit = 2
Total Fault = 5
Hit Ratio = 28.57 %
Fault Ratio = 71.43 %
PS C:\Users\hp\Desktop\CSE 402 Lab>
```

4. Discussion:

This experiment implements the FIFO Page Replacement Algorithm to manage pages in a limited number of memory frames. The algorithm follows the principle that the page loaded earliest is replaced first when all frames are occupied.

Initially, pages 1, 2, and 3 are loaded into the three available frames, producing three page faults. When page 4 arrives, page 1 is removed because it entered the frames first. The subsequent references to pages 2 and 3 produce page hits because both pages are already available in memory. Finally, page 1 causes another page fault and replaces page 2, which is then the oldest page in the frames.

The final result consists of 2 page hits and 5 page faults. The hit ratio is 28.57%, while the fault ratio is 71.43%. These results show how FIFO replaces pages based on their arrival order, without considering how frequently or recently they have been used.

5. Conclusion:

The FIFO Page Replacement Algorithm was successfully implemented using Python with a frame size of three and the reference string 1, 2, 3, 4, 2, 3, 1. The program identified 2 page hits and 5 page faults, resulting in a hit ratio of 28.57% and a fault ratio of 71.43%. The experiment demonstrates how FIFO manages memory by replacing the page that has remained in the frames for the longest time. Although the algorithm is simple to implement and understand, it does not consider page usage frequency or recency when selecting a page for replacement. Therefore, its performance may vary depending on the reference string and frame size.

6. GitHub:

https://github.com/Rizon40/CSE-402-Operating-System-Lab